



⌚ Time: 5 minutes

⌚ Outcome: Calculate and interpret output losses associated with disinflation.

⌚ Difficulty: Intermediate

## Sacrifice Ratio



### THE CORE IDEA

The sacrifice ratio measures cumulative output lost for each percentage-point reduction in inflation. Credibility can speed expectations adjustment, but disinflation costs depend on model assumptions and wage-price adjustment. The preferred path depends on credibility, expectations, adjustment speed, uncertainty, and cumulative costs. The sacrifice ratio measures how much output is lost when inflation is reduced. If expected inflation falls faster, the short-run cost of disinflation can be smaller.



### HOW TO RECOGNIZE IT

- The sacrifice ratio measures how much output is lost when inflation is reduced.
- If expected inflation falls faster, the short-run cost of disinflation can be smaller.
- The sacrifice ratio measures cumulative output loss per percentage-point reduction in inflation.
- A disinflation has a lower sacrifice ratio than another.
- Faster adjustment of expected inflation reduces the output cost of disinflation.



### WATCH OUT

A common mistake is to divide the dollar output loss directly by the inflation change. First express the loss as a percentage of potential output, then divide by the percentage-point reduction in inflation.



### WORKED EXAMPLE: CALCULATING THE SACRIFICE RATIO

Disinflation output losses (billions of dollars)

|            | Potential | Actual | Output loss | Gap / potential |
|------------|-----------|--------|-------------|-----------------|
| Year 1     | 400       | 388    | 12          | 3%              |
| Year 2     | 400       | 388    | 12          | 3%              |
| Total loss | —         | —      | 24          | 6% of one year  |

Each year loses \$12 billion relative to \$400 billion potential output, a 3% gap. Add the two annual gaps: cumulative loss is 6% of one year's potential output. Inflation falls by 3 percentage points, so the sacrifice ratio is  $6 / 3 = 2$ . Keep percentage output losses distinct from percentage-point inflation changes.



### CHECK YOURSELF

Should annual output gaps be added before calculating a multi-year sacrifice ratio?



**READY?**

**Return to the game and master this concept.**